



Question Set

Question 1

Consider an arithmetic-average Asian call. Define the state variables (running average, terminal spot, averaging dates, and strike) and derive the path-dependent payoff and control-variate estimator. State the measure, filtration, and tradability assumptions.

$$A_T = \frac{1}{T} \int_0^T S_t dt, \quad X_T = (A_T - K)^+$$

Question 2

For an arithmetic-average Asian call, explain which probability measure is natural for pricing. Write the density-process logic needed if the starting model is under the physical measure.

Question 3

Identify the discounted tradable or density-adjusted process in an arithmetic-average Asian call. Prove the martingale condition and name one modelling choice that would break it.

Question 4

Using monthly averaging over one year with spot 100, strike 100, and volatility 22 percent, construct a two-step numerical calculation. Show intermediate state values before discounting or aggregating.

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Question 5

For an arithmetic-average Asian call, compute and interpret delta of the average payoff and seasonality of remaining fixings. Explain the hedge sign and the unit of each sensitivity.

Question 6

Suppose the desk observes vanilla volatility input used for path simulation. Write the calibration objective, the weights you would use, and the parameter-identifiability checks.

Question 7

State the terminal and boundary conditions for an arithmetic-average Asian call. Explain how running-average state at maturity and early fixing limits prevents economically impossible prices.

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Question 8

Design a Monte Carlo or lattice estimator for an arithmetic-average Asian call. Specify Monte Carlo with geometric-average control variate, the estimator, the discounting rule, and the standard error.

Question 9

A junior quant implements an arithmetic-average Asian call and gets a stable but wrong answer. Explain how forgetting already fixed observations in the running average creates the error and design a diagnostic test.

Question 10

Write an NumPy algorithm for an arithmetic-average Asian call. Include inputs, outputs, diagnostics, and one benchmark case.

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Question 11

Starting from the payoff definition of an arithmetic-average Asian call, derive a replication, dynamic-programming, or conditional-expectation representation without skipping the economic argument.

Question 12

Create a stressed-market version of monthly averaging over one year with spot 100, strike 100, and volatility 22 percent. Recompute the value or risk metric and explain which term dominates the change.

Question 13

Use a second-order Taylor approximation for an arithmetic-average Asian call. Separate linear exposure, convexity exposure, cross terms if relevant, and residual model risk.

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Question 14

Show how a change of numeraire or change of measure would simplify one expectation in an arithmetic-average Asian call. Identify what becomes a martingale after the change.

Question 15

Give a proof-style answer showing when the wealth process for an arithmetic-average Asian call is self-financing. Explain why predictable trading strategies matter.

Question 16

Market quotes for vanilla volatility input used for path simulation are noisy. Propose a robust calibration workflow with initial guesses, constraints, residual plots, and out-of-sample checks.

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Question 17

For an arithmetic-average Asian call, compare the mathematical boundary condition with the trader's intuition. Give one limiting case where the answer should be obvious.

Question 18

Design a convergence study for Monte Carlo with geometric-average control variate. Specify path counts, time-step refinement, confidence intervals, and the exact quantity plotted.

Question 19

Find one way that forgetting already fixed observations in the running average could pass a superficial unit test. Write a stronger test that would fail for the biased implementation.

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Question 20

Explain an arithmetic-average Asian call and code the core routine. Give the derivation outline, pseudocode, and three numerical sanity checks.